

Class 10: Structural Bioinformatics Pt. 1

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Background

The main repository of high-resolution structural data on biomolecules is called the **Protein Data Bank** (PDB).

PDB Statistics

What is in the PDB in terms of molecule type and structure determination method?

```
library(readr)
pdb<- read_csv("Data Export Summary.csv")
```

Rows: 6 Columns: 9

-- Column specification -----

Delimiter: ","

chr (1): Molecular Type

dbl (4): Integrative, Multiple methods, Neutron, Other

num (4): X-ray, EM, NMR, Total

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
head(pdb)
```

A tibble: 6 x 9

`Molecular Type` <chr>	`X-ray` <dbl>	EM <dbl>	NMR <dbl>	Integrative <dbl>	`Multiple methods` <dbl>	Neutron <dbl>
1 Protein (only)	180758	23111	12813	348	229	84
2 Protein/Oligosacch~	10488	3741	34	8	11	1
3 Protein/NA	9205	6751	287	26	8	0

4 Nucleic acid (only)	3154	250	1578	3	15	3
5 Other	178	27	35	4	0	0
6 Oligosaccharide (o~	11	0	6	0	1	0

i 2 more variables: Other <dbl>, Total <dbl>

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

93.87% of structures in the PDB are solved by X-Ray and Electron Microscopy.

```
x <- sum(pdb$`X-ray`) + sum(pdb$`EM`)
y <- sum(pdb$Total)
x/y * 100
```

[1] 93.86623

Q2: What proportion of structures in the PDB are protein?

The total number of protein sequences in UniProt is 202556314. There are 217375 in the dataset. Leading to a ~0.1%.

```
pdb$Total[1]/202556314 * 100
```

[1] 0.1073158

Key-Point: We have a very, very small structural coverage of known proteins (~0.1%). Most structures we know are about (~80%) come from one method (X-Ray Crystallography).

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

After typing HIV into the search box in PDB, I saw that there were 4998 Structures that match the query.

Visualizing PDB Data with Mol-Star

Main stand alone web version with all features is at <https://molstar.org/viewer/>.



Figure 1: The HIV-Protease Enzyme is a homodimer with two chains.

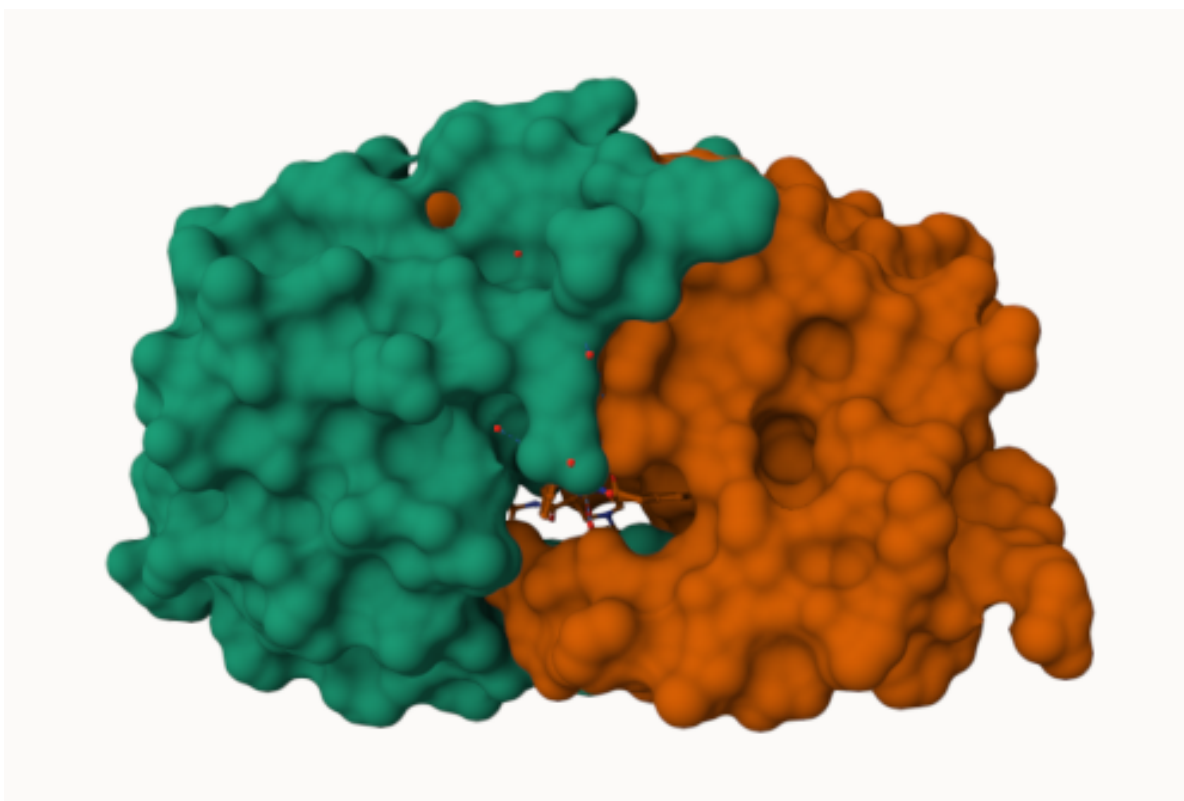


Figure 2: Surface display showing the binding cleft site for the inhibitor (drug) molecule.



Figure 3: Spacefill display of catalytic ASP25 amino acids and key binding site water molecule

GETting started with the Bio3D package

Bio3D is an R package from CRAN for structural bioinformatics

Q7

Q8

Q9

```
library(bio3d)
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

pdb

Call: read.pdb(file = "1hsg")

Total Models#: 1

Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)

Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172 (residues: 128)

Non-protein/nucleic resid values: [HOH (127), MK1 (1)]

Protein sequence:

PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF

+ attr: atom, xyz, seqres, helix, sheet,
calpha, remark, call

pdb\$atom

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1
7	ATOM	7	CD	<NA>	PRO	A	1	<NA>	28.778	39.015	7.019	1
8	ATOM	8	N	<NA>	GLN	A	2	<NA>	30.607	37.334	3.305	1
9	ATOM	9	CA	<NA>	GLN	A	2	<NA>	30.158	36.492	2.199	1
10	ATOM	10	C	<NA>	GLN	A	2	<NA>	30.298	35.041	2.643	1
11	ATOM	11	O	<NA>	GLN	A	2	<NA>	31.401	34.494	2.763	1
12	ATOM	12	CB	<NA>	GLN	A	2	<NA>	30.970	36.738	0.926	1
13	ATOM	13	CG	<NA>	GLN	A	2	<NA>	30.625	35.783	-	

0.201 1

14	ATOM	14	CD <NA>	GLN	A	2	<NA>	31.184	36.217	-
1.549	1									
15	ATOM	15	OE1 <NA>	GLN	A	2	<NA>	32.006	35.518	-
2.156	1									
16	ATOM	16	NE2 <NA>	GLN	A	2	<NA>	30.684	37.339	-
2.061	1									
17	ATOM	17	N <NA>	ILE	A	3	<NA>	29.160	34.436	2.919 1
18	ATOM	18	CA <NA>	ILE	A	3	<NA>	29.123	33.098	3.397 1
19	ATOM	19	C <NA>	ILE	A	3	<NA>	28.968	32.155	2.198 1
20	ATOM	20	O <NA>	ILE	A	3	<NA>	28.088	32.330	1.368 1
21	ATOM	21	CB <NA>	ILE	A	3	<NA>	27.977	32.995	4.409 1
22	ATOM	22	CG1 <NA>	ILE	A	3	<NA>	28.341	33.820	5.652 1
23	ATOM	23	CG2 <NA>	ILE	A	3	<NA>	27.692	31.548	4.745 1
24	ATOM	24	CD1 <NA>	ILE	A	3	<NA>	27.264	33.884	6.696 1
25	ATOM	25	N <NA>	THR	A	4	<NA>	29.891	31.210	2.066 1
26	ATOM	26	CA <NA>	THR	A	4	<NA>	29.774	30.143	1.062 1
27	ATOM	27	C <NA>	THR	A	4	<NA>	28.986	28.975	1.658 1
28	ATOM	28	O <NA>	THR	A	4	<NA>	28.690	28.948	2.875 1
29	ATOM	29	CB <NA>	THR	A	4	<NA>	31.165	29.618	0.634 1
30	ATOM	30	OG1 <NA>	THR	A	4	<NA>	31.866	29.209	1.815 1
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0.085	1									
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33	ATOM	33	CA <NA>	LEU	A	5	<NA>	27.644	27.003	1.144 1
34	ATOM	34	C <NA>	LEU	A	5	<NA>	28.204	25.559	1.071 1
35	ATOM	35	O <NA>	LEU	A	5	<NA>	27.446	24.583	0.969 1
36	ATOM	36	CB <NA>	LEU	A	5	<NA>	26.411	27.139	0.226 1
37	ATOM	37	CG <NA>	LEU	A	5	<NA>	25.676	28.479	0.352 1
38	ATOM	38	CD1 <NA>	LEU	A	5	<NA>	24.624	28.624	-
0.753	1									
39	ATOM	39	CD2 <NA>	LEU	A	5	<NA>	25.088	28.590	1.745 1
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42	ATOM	42	C <NA>	TRP	A	6	<NA>	29.837	23.488	2.611 1
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0.189	1									
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0.345	1									
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1.478	1									
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1.655	1												
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2.369	1												
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1.986	1												
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4.169	1												
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66	ATOM	66	O <NA>	ARG	A	8	<NA>	26.947	26.291	7.775	1		
67	ATOM	67	CB <NA>	ARG	A	8	<NA>	25.087	23.849	7.776	1		
68	ATOM	68	CG <NA>	ARG	A	8	<NA>	24.646	22.409	7.505	1		
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75	ATOM	75	CA <NA>	PRO	A	9	<NA>	25.491	28.352	6.938	1		
76	ATOM	76	C <NA>	PRO	A	9	<NA>	25.127	28.763	8.364	1		
77	ATOM	77	O <NA>	PRO	A	9	<NA>	24.136	29.472	8.578	1		
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79	ATOM	79	CG <NA>	PRO	A	9	<NA>	23.625	28.254	5.407	1		
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81	ATOM	81	N <NA>	LEU	A	10	<NA>	25.905	28.285	9.330	1		
82	ATOM	82	CA <NA>	LEU	A	10	<NA>	25.653	28.510	10.750	1		
83	ATOM	83	C <NA>	LEU	A	10	<NA>	26.383	29.770	11.208	1		
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86	ATOM	86	CG <NA>	LEU	A	10	<NA>	25.161	26.082	11.544	1
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91	ATOM	91	C <NA>	VAL	A	11	<NA>	25.818	31.957	13.955	1
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93	ATOM	93	CB <NA>	VAL	A	11	<NA>	25.824	33.131	11.791	1
94	ATOM	94	CG1 <NA>	VAL	A	11	<NA>	26.270	33.089	10.323	1
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96	ATOM	96	N <NA>	THR	A	12	<NA>	26.397	32.913	14.700	1
97	ATOM	97	CA <NA>	THR	A	12	<NA>	26.001	33.143	16.102	1
98	ATOM	98	C <NA>	THR	A	12	<NA>	24.915	34.200	16.204	1
99	ATOM	99	O <NA>	THR	A	12	<NA>	25.010	35.279	15.610	1
100	ATOM	100	CB <NA>	THR	A	12	<NA>	27.201	33.565	16.998	1
101	ATOM	101	OG1 <NA>	THR	A	12	<NA>	28.330	32.709	16.751	1
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104	ATOM	104	CA <NA>	ILE	A	13	<NA>	22.842	34.875	17.206	1
105	ATOM	105	C <NA>	ILE	A	13	<NA>	22.770	35.114	18.707	1
106	ATOM	106	O <NA>	ILE	A	13	<NA>	23.328	34.363	19.500	1
107	ATOM	107	CB <NA>	ILE	A	13	<NA>	21.413	34.460	16.661	1
108	ATOM	108	CG1 <NA>	ILE	A	13	<NA>	20.878	33.229	17.431	1
109	ATOM	109	CG2 <NA>	ILE	A	13	<NA>	21.510	34.194	15.162	1
110	ATOM	110	CD1 <NA>	ILE	A	13	<NA>	19.353	33.201	17.603	1
111	ATOM	111	N <NA>	LYS	A	14	<NA>	22.106	36.199	19.087	1
112	ATOM	112	CA <NA>	LYS	A	14	<NA>	21.894	36.545	20.492	1
113	ATOM	113	C <NA>	LYS	A	14	<NA>	20.442	36.943	20.615	1
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115	ATOM	115	CB <NA>	LYS	A	14	<NA>	22.777	37.724	20.896	1
116	ATOM	116	CG <NA>	LYS	A	14	<NA>	22.727	38.056	22.383	1
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118	ATOM	118	CE <NA>	LYS	A	14	<NA>	24.814	39.490	22.755	1
119	ATOM	119	NZ <NA>	LYS	A	14	<NA>	25.394	40.891	22.572	1
120	ATOM	120	N <NA>	ILE	A	15	<NA>	19.739	36.267	21.512	1
121	ATOM	121	CA <NA>	ILE	A	15	<NA>	18.345	36.563	21.813	1
122	ATOM	122	C <NA>	ILE	A	15	<NA>	18.224	36.327	23.316	1
123	ATOM	123	O <NA>	ILE	A	15	<NA>	18.886	35.449	23.864	1
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126	ATOM	126	CG2 <NA>	ILE	A	15	<NA>	17.745	34.137	21.266	1
127	ATOM	127	CD1 <NA>	ILE	A	15	<NA>	14.929	35.116	20.526	1
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129	ATOM	129	CA <NA>	GLY	A	16	<NA>	17.356	36.968	25.459	1
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141	ATOM	141	CG <NA>	GLN	A	18	<NA>	19.599	33.116	26.176	1
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161	ATOM	161	NZ <NA>	LYS	A	20	<NA>	18.467	28.892	22.571	1
162	ATOM	162	N <NA>	GLU	A	21	<NA>	23.725	29.250	18.342	1
163	ATOM	163	CA <NA>	GLU	A	21	<NA>	24.053	29.117	16.931	1
164	ATOM	164	C <NA>	GLU	A	21	<NA>	22.822	28.761	16.150	1
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167	ATOM	167	CG <NA>	GLU	A	21	<NA>	25.035	26.716	17.168	1
168	ATOM	168	CD <NA>	GLU	A	21	<NA>	25.878	25.743	16.334	1
169	ATOM	169	OE1 <NA>	GLU	A	21	<NA>	27.022	26.130	15.972	1
170	ATOM	170	OE2 <NA>	GLU	A	21	<NA>	25.379	24.639	15.983	1
171	ATOM	171	N <NA>	ALA	A	22	<NA>	22.778	29.268	14.927	1

172	ATOM	172	CA <NA>	ALA	A	22	<NA>	21.553	29.189	14.165	1
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175	ATOM	175	CB <NA>	ALA	A	22	<NA>	20.625	30.359	14.524	1
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1608	HETATM	1610	0	<NA>	HOH	B	308	<NA>	11.720	21.289	7.190	1
1609	HETATM	1611	0	<NA>	HOH	B	312	<NA>	14.097	5.111	-11.638	1
1610	HETATM	1612	0	<NA>	HOH	B	313	<NA>	20.998	21.834	6.561	1
1611	HETATM	1613	0	<NA>	HOH	B	316	<NA>	22.659	14.583	-	
2.196												1
1612	HETATM	1614	0	<NA>	HOH	B	317	<NA>	28.724	15.629	-11.660	1
1613	HETATM	1615	0	<NA>	HOH	B	318	<NA>	16.539	45.207	0.079	1
1614	HETATM	1616	0	<NA>	HOH	B	319	<NA>	23.678	14.931	2.680	1
1615	HETATM	1617	0	<NA>	HOH	B	321	<NA>	20.718	15.976	-	
3.657												1
1616	HETATM	1618	0	<NA>	HOH	B	323	<NA>	31.249	26.796	-	
9.595												1
1617	HETATM	1619	0	<NA>	HOH	B	326	<NA>	28.813	28.445	-	
2.106												1

1618 HETATM	1620	0 <NA>	HOH	B	333	<NA>	12.251	39.551	-	
2.672	1									
1619 HETATM	1621	0 <NA>	HOH	B	334	<NA>	25.465	12.592	-	
8.670	1									
1620 HETATM	1622	0 <NA>	HOH	B	338	<NA>	12.998	36.205	-	
3.972	1									
1621 HETATM	1623	0 <NA>	HOH	B	339	<NA>	17.541	17.060	-17.194	1
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4.866	1									
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9.801	1									
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3.635	1									
1626 HETATM	1628	0 <NA>	HOH	B	349	<NA>	19.414	7.026	4.428	1
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1.322	1									
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6.155	1									
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5.102	1									
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7.529	1									
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1637 HETATM	1639	0 <NA>	HOH	B	367	<NA>	25.658	18.970	0.428	1
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7.014	1									
1639 HETATM	1641	0 <NA>	HOH	B	370	<NA>	13.975	15.741	12.070	1
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5.608 1

1645 HETATM	1647	0 <NA>	HOH	B	381	<NA>	25.810	26.789	-19.106	1
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1648 HETATM	1650	0 <NA>	HOH	B	388	<NA>	23.926	45.612	-	

4.998 1

1649 HETATM	1651	0 <NA>	HOH	B	390	<NA>	25.300	17.493	3.076	1
1650 HETATM	1652	0 <NA>	HOH	B	392	<NA>	6.618	28.079	-	

3.427 1

1651 HETATM	1653	0 <NA>	HOH	B	393	<NA>	19.795	13.651	-16.606	1
1652 HETATM	1654	0 <NA>	HOH	B	395	<NA>	7.202	9.982	-	

4.103 1

1653 HETATM	1655	0 <NA>	HOH	B	400	<NA>	8.474	34.203	-	
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4.893 1

1654 HETATM	1656	0 <NA>	HOH	B	405	<NA>	16.659	15.866	11.446	1
1655 HETATM	1657	0 <NA>	HOH	B	410	<NA>	26.400	10.057	-	

3.287 1

1656 HETATM	1658	0 <NA>	HOH	B	414	<NA>	9.503	3.489	-	
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4.419 1

1657 HETATM	1659	0 <NA>	HOH	B	419	<NA>	15.438	12.973	-18.484	1
1658 HETATM	1660	0 <NA>	HOH	B	425	<NA>	11.428	19.956	-24.551	1
1659 HETATM	1661	0 <NA>	HOH	B	430	<NA>	18.725	43.171	-	

5.575 1

1660 HETATM	1662	0 <NA>	HOH	B	436	<NA>	32.141	29.620	-	
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8.580 1

1661 HETATM	1663	0 <NA>	HOH	B	443	<NA>	8.811	13.667	-20.256	1
1662 HETATM	1664	0 <NA>	HOH	B	444	<NA>	4.071	26.169	-	

0.230 1

1663 HETATM	1665	0 <NA>	HOH	B	461	<NA>	11.425	44.636	-	
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3.033 1

1664 HETATM	1666	0 <NA>	HOH	B	469	<NA>	6.902	23.686	-10.066	1
1665 HETATM	1667	0 <NA>	HOH	B	471	<NA>	5.749	25.785	-19.792	1
1666 HETATM	1668	0 <NA>	HOH	B	500	<NA>	25.592	16.404	-	

5.805 1

1667 HETATM	1669	0 <NA>	HOH	B	502	<NA>	4.040	15.516	-	
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7.200 1

1668 HETATM	1670	0 <NA>	HOH	B	505	<NA>	28.640	34.232	-	
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5.637 1

1669 HETATM	1671	0 <NA>	HOH	B	506	<NA>	8.979	11.173	11.112	1
1670 HETATM	1672	0 <NA>	HOH	B	509	<NA>	19.882	3.986	-18.136	1
1671 HETATM	1673	0 <NA>	HOH	B	514	<NA>	27.409	15.355	2.200	1
1672 HETATM	1674	0 <NA>	HOH	B	515	<NA>	17.222	39.766	-23.774	1
1673 HETATM	1675	0 <NA>	HOH	B	517	<NA>	28.742	24.158	-16.641	1

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1676	HETATM	1678	O	<NA>	HOH	B	531	<NA>	18.192	8.914	11.344	1
1677	HETATM	1679	O	<NA>	HOH	B	532	<NA>	19.507	45.215	1.709	1
1678	HETATM	1680	O	<NA>	HOH	B	548	<NA>	1.442	14.700	-	
6.128 1												
1679	HETATM	1681	O	<NA>	HOH	B	549	<NA>	19.908	8.718	-19.215	1
1680	HETATM	1682	O	<NA>	HOH	B	556	<NA>	21.499	44.884	-	
1.280 1												
1681	HETATM	1683	O	<NA>	HOH	B	564	<NA>	10.031	8.593	-22.052	1
1682	HETATM	1684	O	<NA>	HOH	B	568	<NA>	2.817	28.133	2.191	1
1683	HETATM	1685	O	<NA>	HOH	B	591	<NA>	15.835	40.105	-	
5.971 1												
1684	HETATM	1686	O	<NA>	HOH	B	595	<NA>	4.515	36.451	-	
4.499 1												
1685	HETATM	1687	O	<NA>	HOH	B	613	<NA>	24.127	-10.994	-	
0.982 1												
1686	HETATM	1688	O	<NA>	HOH	B	617	<NA>	30.112	17.912	-	
4.791 1												

b segid elesy charge

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1653	63.64	<NA>	0	<NA>
1654	42.47	<NA>	0	<NA>
1655	65.50	<NA>	0	<NA>
1656	65.50	<NA>	0	<NA>
1657	73.55	<NA>	0	<NA>

1658	63.48	<NA>	0	<NA>
1659	52.97	<NA>	0	<NA>
1660	72.75	<NA>	0	<NA>
1661	75.75	<NA>	0	<NA>
1662	38.25	<NA>	0	<NA>
1663	68.43	<NA>	0	<NA>
1664	54.20	<NA>	0	<NA>
1665	63.96	<NA>	0	<NA>
1666	23.98	<NA>	0	<NA>
1667	52.93	<NA>	0	<NA>
1668	58.06	<NA>	0	<NA>
1669	64.79	<NA>	0	<NA>
1670	55.54	<NA>	0	<NA>
1671	61.69	<NA>	0	<NA>
1672	69.12	<NA>	0	<NA>
1673	78.93	<NA>	0	<NA>
1674	71.37	<NA>	0	<NA>
1675	78.14	<NA>	0	<NA>
1676	54.05	<NA>	0	<NA>
1677	72.78	<NA>	0	<NA>
1678	58.40	<NA>	0	<NA>
1679	58.78	<NA>	0	<NA>
1680	68.40	<NA>	0	<NA>
1681	64.90	<NA>	0	<NA>
1682	67.95	<NA>	0	<NA>
1683	53.68	<NA>	0	<NA>
1684	49.41	<NA>	0	<NA>
1685	64.49	<NA>	0	<NA>
1686	54.09	<NA>	0	<NA>

```
attributes(pdb)
```

```
$names
```

```
[1] "atom"    "xyz"     "seqres"  "helix"   "sheet"   "calpha"  "remark"  "call"
```

```
$class
```

```
[1] "pdb" "sse"
```

There are lots of functions that can work with these `pdb` objects.

```
head( pdbseq(pdb))
```

```
  1   2   3   4   5   6  
"P" "Q" "I" "T" "L" "W"
```

We can have a quick interactive view of any of these `pdb` objects:

```
#install.packages("pak")  
  
#pak::pkg_install(c("bioboot/bio3dview",  
                    #"NGLVieweR",  
                    #"bioc::msa")  
                  #)
```

```
library(bio3dview)
```

```
#view.pdb(pdb)
```

Predict the flexibility of a given structure

Let's do a Normal Mode Analysis (NMA) to predict the flexibility of a given `pdb` object:

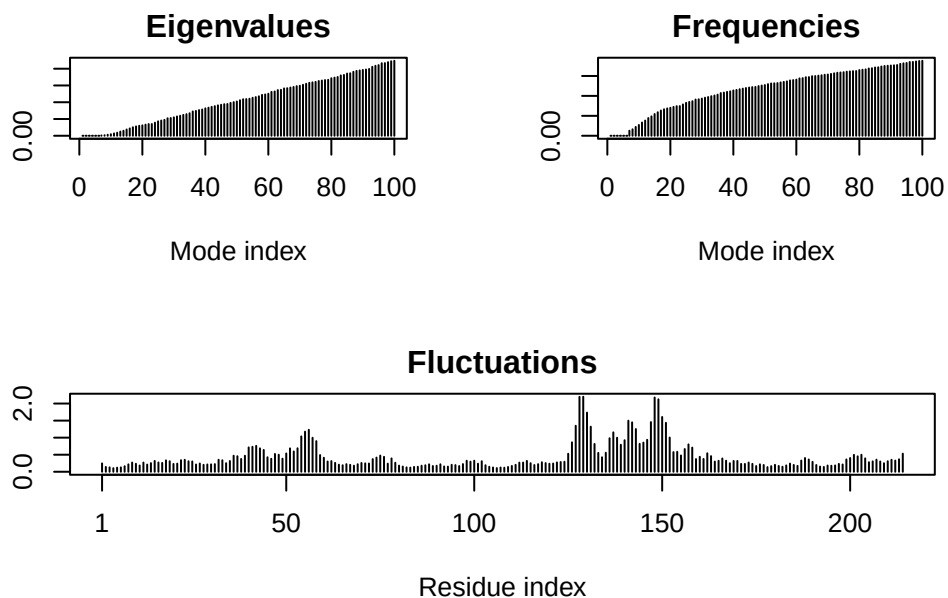
```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, `rm.alt=TRUE`

```
m <- nma(adk)
```

```
Building Hessian...      Done in 0.06 seconds.  
Diagonalizing Hessian... Done in 0.34 seconds.
```

```
plot(m)
```

View the results with an interactive structure view

```
#view.nma(m)
```

```
mktrj(m, file="nma.pdb")
```

Q. Create a custom view highlighting the active site ASP (resno=25), the two chains (in your choice of colors) and the ligand all on a custom color background.

```
#library(NGLViewer)
#active.site <- atom.select(pdb, resno=25)

#view.pdb(pdb,
  #cols = c("red", "blue"),
  #highlight = active.site,
  #highlight.style = "spacefill",
  #backgroundColor = "pink") |>
#setRock()
```

Comparative analysis of the ADK family

Our first step is find a sequence for this family. We will use the database ID “1ake_A” here:

```
id <- "1ake_A"

aa <- get.seq(id)
```

Warning in get.seq(id): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

```
aa
```

```

      1      .      .      .      .      .      60
pdb|1AKE|A  MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAAVKSGSELGKQAKDIMDAGKLV
      1      .      .      .      .      .      60

      61      .      .      .      .      .      120
pdb|1AKE|A  DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
      61      .      .      .      .      .      120

      121      .      .      .      .      .      180
pdb|1AKE|A  VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTPALIG
      121      .      .      .      .      .      180

      181      .      .      .      214
pdb|1AKE|A  YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
      181      .      .      .      214
```

```
Call:
  read.fasta(file = outfile)
```

```
Class:
  fasta
```

```
Alignment dimensions:
  1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

```
+ attr: id, ali, call
```

Search for related sequences in the database

```
#blast <- blast.pdb(aa)
```

```
#head(blast$hit.tbl)
```

```
#hits <- plot(blast)
```

```
#hits$pdb.id
```

```
#f <- get.pdb(hits$pdb.id, path="pdbs", split = T, gzip = T)
```

```
#pdbs <- pdbaln(f, fit= TRUE, exefile = "msa")  
#pdbs
```

```
#view.pdbs(pdbs)
```

PCA of all this structural data (x, y, and z atom coordinates):

```
#pc <- pca(pdbs)  
#plot(pc)
```

```
#plot(pc, 1:2)
```

Interactive view of the PC1 captured structural

```
#view.pca(pc)
```

```
#mktrj(pc, file="pca.pdb")
```